

# Grade 11 – Term 1

## C5 Practice Activity 5 with Solutions

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### Evaluated criterion

- C6.** Estimates interval probabilities using relative frequencies from generated samples and compares them with theoretical probabilities.
- C7.** Calculates sample means and variances for different sample sizes and compares them with the theoretical mean and variance.

## 1 Introduction

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### Problems:

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- 1.1. Generated samples, sample sizes, and histograms
  - 1.2. Continuous distributions, theoretical and empirical probabilities
  - 1.3. Sample mean and variance from generated data
- 

Criterion C6 focuses on estimating interval probabilities from generated samples and comparing these empirical estimates with the theoretical probabilities predicted by a continuous probability distribution.

The comparison can be understood through two representations of the same distribution. The probability density function (PDF) represents the theoretical model as a smooth curve. A histogram represents the observations obtained from a finite generated sample. The theoretical probability of an interval is determined by the area under the PDF, while the empirical probability is estimated from the relative frequency of observations in the corresponding interval.

The normal distribution provides a useful example because its theoretical density has a smooth bell-shaped form, while a histogram generated from the same distribution produces a finite and irregular approximation to that shape.

### 1.1 Real-world data, probability distributions, and histograms

Real-world data often show patterns of variation that can be described using probability distributions. A theoretical distribution provides a model for how values of a variable may be distributed, while a dataset contains the observations

actually collected or generated from that process.

A dataset with  $n$  observations can be written as

$$x_1, x_2, \dots, x_n.$$

The observations can be organized into a histogram, which groups values into intervals and shows their frequencies. The histogram provides an empirical description of the data that can be compared with the theoretical probability distribution.

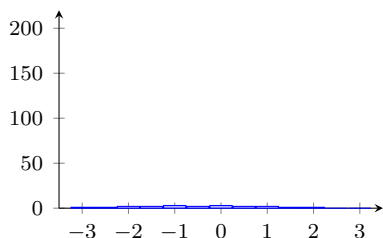


Figure 1: Count histogram from a sample of  $n = 20$  observations.

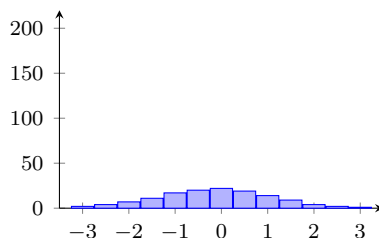


Figure 2: Count histogram from a sample of  $n = 100$  observations.

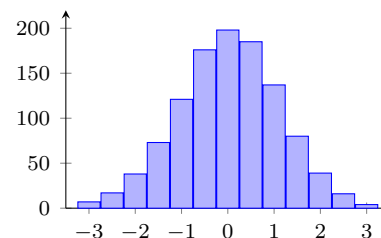


Figure 3: Count histogram from a sample of  $n = 1000$  observations.

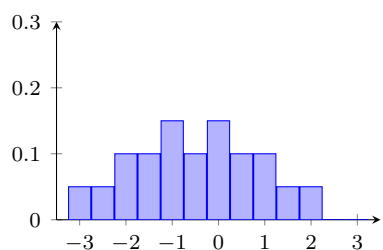


Figure 4: Histogram from a sample of  $n = 20$  observations.

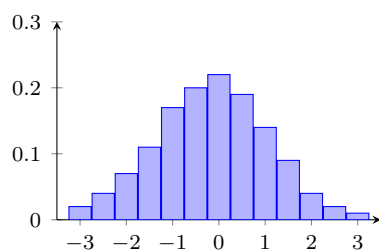


Figure 5: Histogram from a sample of  $n = 100$  observations.

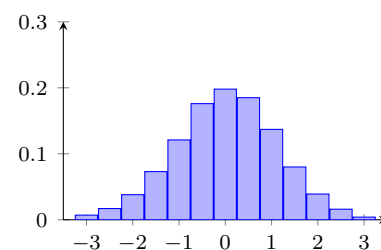


Figure 6: Histogram from a sample of  $n = 1000$  observations.

Different samples from the same process can produce different histograms. Larger samples generally provide a more stable representation of the underlying distribution because random variation has less influence on the overall shape.

The theoretical distribution and the observed data can also be compared numerically. The theoretical distribution has mean  $\mu$  and variance  $\sigma^2$ , while the sample has mean  $\bar{x}$  and variance  $s^2$ :

| Theoretical model               | Observed data                       |
|---------------------------------|-------------------------------------|
| Probability distribution        | Observations $x_1, x_2, \dots, x_n$ |
| Theoretical mean $\mu$          | Sample mean $\bar{x}$               |
| Theoretical variance $\sigma^2$ | Sample variance $s^2$               |

As the sample size increases, the sample statistics generally become more stable and tend to approximate the corresponding theoretical parameters:

$$\bar{x} \approx \mu, \quad s^2 \approx \sigma^2.$$

Thus, the relationship can be summarized as

Real-world process  $\longrightarrow$  Observed data  $\longrightarrow$  Histogram and sample statistics  $\longleftrightarrow$  Probability distribution

The following subsection presents the equations used to calculate the sample mean and variance from the data.

## 1.2 Continuous distributions, theoretical and empirical probabilities

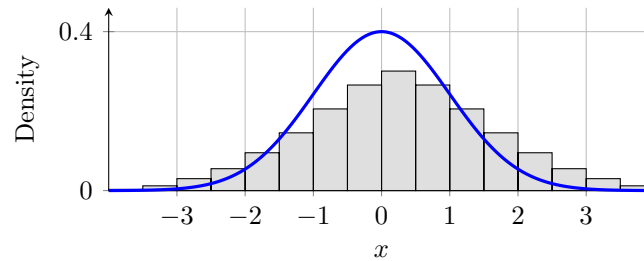


Figure 7: Theoretical normal probability density function and a histogram from a sample generated from the same distribution. The smooth curve represents the theoretical model, while the bars represent the observed sample.

Continuous probability distributions provide theoretical models for continuous random variables. For an interval  $[a, b]$ , the theoretical probability is determined by the area under the probability density function:

$$P(a \leq X \leq b) = \int_a^b f(x) dx.$$

When a finite sample is generated from the distribution, the corresponding probability can instead be estimated from the observations. If  $k$  of the  $n$  observations fall within the interval, the empirical probability is given by the relative frequency

$$\hat{P}(a \leq X \leq b) = \frac{k}{n}, \quad k = \#\{x_i : a \leq x_i \leq b\}.$$

The distinction between the two probabilities can be summarized as follows:

| Theoretical probability                 | Empirical probability                    |
|-----------------------------------------|------------------------------------------|
| Based on the probability distribution   | Based on a generated sample              |
| Area under the PDF                      | Relative frequency                       |
| $P(a \leq X \leq b) = \int_a^b f(x) dx$ | $\hat{P}(a \leq X \leq b) = \frac{k}{n}$ |

The following pair of graphs illustrates the two calculations separately:

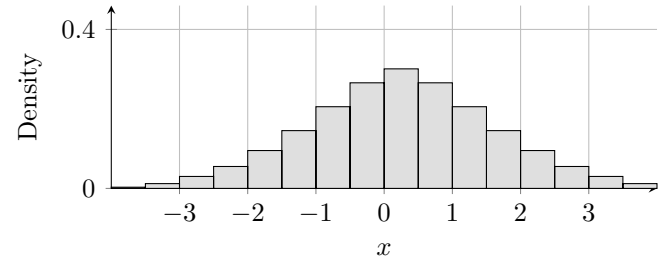
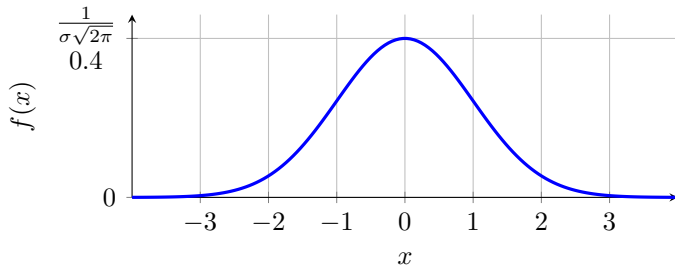


Figure 8: Theoretical probability density function of a normal distribution. The area under the curve over an interval represents the theoretical probability of that interval.

Figure 9: Histogram of a finite sample generated from the same distribution. The bars show the observed frequencies within the histogram bins.

The histogram is useful because it connects the generated observations with the empirical calculation. The bars show where the observations occur, and the observations within the selected interval determine  $k$ .

For all the continuous distributions used in this criterion, the method for calculating the empirical probability is the same:

$$\hat{P} = \frac{\text{number of observations in the interval}}{\text{total number of observations}}.$$

The method for calculating the theoretical probability depends on the probability density function of the distribution. The key comparison is therefore between the area represented by the theoretical PDF and the proportion of observations represented by the histogram:

$$\text{PDF area} \longleftrightarrow \text{histogram relative frequency}$$

For a sufficiently large generated sample, these two quantities should generally be close, while differences are expected because the histogram is based on a finite random sample.

The difference between the empirical and theoretical probabilities is

$$\text{Difference} = \hat{P}(a \leq X \leq b) - P(a \leq X \leq b).$$

The absolute difference is

$$\text{Absolute difference} = \left| \hat{P}(a \leq X \leq b) - P(a \leq X \leq b) \right|.$$

When the theoretical probability is nonzero, the percentage error is

$$\text{Percentage error} = \frac{\left| \hat{P}(a \leq X \leq b) - P(a \leq X \leq b) \right|}{P(a \leq X \leq b)} \times 100\%.$$

These quantities provide a common reference for comparing theoretical probabilities obtained from a continuous distribution with empirical probabilities obtained from generated samples.



### 1.3 Sample mean and variance from generated data

A theoretical probability distribution describes a model, while a finite generated sample provides observations from that model. The theoretical mean and variance are denoted by  $\mu$  and  $\sigma^2$ , while the corresponding sample statistics are  $\bar{x}$  and  $s^2$ .

| Theoretical model           | Generated sample              |
|-----------------------------|-------------------------------|
| Probability distribution    | Finite dataset                |
| Mean $\mu$                  | Sample mean $\bar{x}$         |
| Variance $\sigma^2$         | Sample variance $s^2$         |
| Standard deviation $\sigma$ | Sample standard deviation $s$ |

For ungrouped observations  $x_1, x_2, \dots, x_n$ , the sample mean, variance, and standard deviation are

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i, \quad s^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2, \quad s = \sqrt{s^2}.$$

The mean describes the center of the observed data, while the variance measures its variability around  $\bar{x}$ .

When observations are grouped into intervals, let  $f_i$  be the frequency of class  $i$  and  $m_i$  its midpoint:

$$m_i = \frac{a_i + b_i}{2}, \quad n = \sum_i f_i.$$

The grouped-data mean and variance are

$$\bar{x} = \frac{\sum_i f_i m_i}{n}, \quad s^2 = \frac{\sum_i f_i (m_i - \bar{x})^2}{n-1}.$$

Grouped calculations are approximate because all observations within an interval are represented by its midpoint.

| Ungrouped data                                 | Grouped data                                     |
|------------------------------------------------|--------------------------------------------------|
| $x_1, x_2, \dots, x_n$                         | Intervals and frequencies                        |
| Exact observations                             | Class midpoints $m_i$                            |
| $\bar{x} = \frac{1}{n} \sum_i x_i$             | $\bar{x} = \frac{\sum_i f_i m_i}{n}$             |
| $s^2 = \frac{1}{n-1} \sum_i (x_i - \bar{x})^2$ | $s^2 = \frac{\sum_i f_i (m_i - \bar{x})^2}{n-1}$ |

For a finite random sample generated from a theoretical distribution,

$$\bar{x} \approx \mu, \quad s^2 \approx \sigma^2.$$

These are approximations: different samples can produce different values. As the sample size increases, the sample statistics generally become more stable and tend toward the corresponding theoretical values.

2 Samples, Empirical Probabilities, and Sample Statistics

| Problems: |
|-----------|
| ??.       |
| 2.2.      |
| 2.3.      |
| 2.4.      |
| 2.5.      |
| 2.6.      |

2.1 Investment returns

A financial analyst records the percentage return of an investment over 20 randomly selected days. The observed returns are

|     |     |     |     |     |     |     |     |     |     |
|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|
| 2.1 | 3.4 | 1.8 | 4.2 | 2.7 | 3.1 | 1.5 | 2.9 | 4.0 | 2.4 |
| 3.6 | 1.9 | 2.2 | 3.8 | 2.6 | 1.7 | 3.3 | 2.5 | 4.1 | 3.0 |

|                                                                                                   |                                                                                  |
|---------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------|
| The returns are modeled theoretically by a normal distribution with theoretical mean and variance | The theoretical probability that a daily return is between 2% and 4% is given as |
| $\mu = 3, \quad \sigma^2 = 1.$                                                                    | $P(2 \leq X \leq 4) = 0.6827.$                                                   |

Question.

- C6 Calculate the empirical probability that the observed return is between 2% and 4%, calculate the absolute error and percentage error with respect to the theoretical probability, and compare the two probabilities.
- C7 Calculate the sample mean and sample variance of the observed returns, and calculate the absolute and percentage errors of these estimates with respect to the theoretical mean and variance.

C6 Solution

Define the variable and sample

$X = \text{daily percentage return of the investment}$

The sample contains  $n = 20$  observations. The analyst wants to estimate the probability that a daily return is between 2% and 4%, i.e.  $P(2 \leq X \leq 4)$ .

From the sample, there are  $k = 13$  observations in this interval. Therefore, the empirical probability is

$$\hat{P}(2 \leq X \leq 4) = \frac{k}{n} = \frac{13}{20} = 0.65$$

The theoretical probability is

$$P(2 \leq X \leq 4) = 0.6827$$

The absolute error is

$$E = \left| \hat{P} - P \right| = |0.65 - 0.6827| = 0.0327$$

The percentage error is calculated relative to the theoretical probability:

$$E_{\%} = \frac{\left| \hat{P} - P \right|}{P} \times 100 = \frac{0.0327}{0.6827} \times 100 \approx 4.79\%$$

## C7 Solution

The sample mean is calculated as

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i. \quad \longrightarrow \quad \bar{x} = \frac{56.8}{20} = 2.84$$

The theoretical mean is  $\mu = 3$ . Therefore,

The absolute error in the sample mean is

$$E_{\bar{x}} = |\bar{x} - \mu| = |2.84 - 3| = 0.16$$

The percentage error is

$$E_{\bar{x},\%} = \frac{|\bar{x} - \mu|}{|\mu|} \times 100 = \frac{0.16}{3} \times 100 \approx 5.33\%$$

The sample variance is calculated using

$$s^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2$$

For the observed returns,

$$s^2 = 0.7004$$

The theoretical variance is

$$\sigma^2 = 1.$$

Therefore, the absolute error in the sample variance is

$$E_{s^2} = |s^2 - \sigma^2| = |0.7004 - 1| = 0.2996.$$

The percentage error is

$$E_{s^2,\%} = \frac{|s^2 - \sigma^2|}{\sigma^2} \times 100 = \frac{0.2996}{1} \times 100 \approx 29.96\%.$$

Thus, the sample mean is 2.84, with an absolute error of 0.16 and a percentage error of approximately 5.33%. The sample variance is 0.7004, with an absolute error of approximately 0.2996 and a percentage error of approximately 29.96%.

## 2.2 Daily energy consumption

A company records the daily electricity consumption of a small production facility. The observed values, measured in MWh, are given with their frequencies:

|       |   |   |    |    |    |    |    |    |
|-------|---|---|----|----|----|----|----|----|
| $x_i$ | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 |
| $f_i$ | 1 | 2 | 3  | 4  | 5  | 4  | 3  | 2  |

The daily consumption is modeled theoretically by a uniform distribution on the interval  $[8, 15]$ .

The theoretical probability that consumption is between 10 MWh and 13 MWh is given as

$$P(10 \leq X \leq 13) = \frac{3}{7} \approx 0.4286.$$

**Question.**

**C4** Calculate the empirical probability that daily consumption is between 10 MWh and 13 MWh, calculate the absolute error with respect to the theoretical probability, and compare the two probabilities.

**Define the variable and frequency distribution**

$X$  = daily electricity consumption, in MWh

The total sample size is

$$n = \sum_i f_i = 1 + 2 + 3 + 4 + 5 + 4 + 3 + 2 = 24.$$

**C6.1 - Find and compare probabilities**

The required interval is

$$10 \leq X \leq 13.$$

The corresponding frequencies are

$$3 + 4 + 5 + 4 = 16.$$

Therefore, the empirical probability is

$$\hat{P}(10 \leq X \leq 13) = \frac{16}{24} = 0.6667.$$

The theoretical probability is

$$P(10 \leq X \leq 13) = \frac{3}{7} \approx 0.4286.$$

The absolute error is

$$E = |0.6667 - 0.4286| = 0.2381.$$

Thus,

$$\hat{P} = 0.6667, \quad P = 0.4286, \quad E = 0.2381$$

The empirical probability differs from the theoretical probability. The difference reflects the finite sample and the particular observations obtained.

## 2.3 Delivery times

A delivery company records the time required to complete deliveries, measured in minutes. The observations are grouped into intervals:

| Delivery time (min) | Frequency |
|---------------------|-----------|
| 10–20               | 6         |
| 20–30               | 14        |
| 30–40               | 18        |
| 40–50               | 9         |
| 50–60               | 3         |

The delivery time is modeled theoretically by an exponential distribution. The theoretical probability that a delivery takes between 20 and 40 minutes is given as

$$P(20 \leq X \leq 40) = 0.4866.$$

### Question.

**C4** Estimate the empirical probability that a delivery takes between 20 and 40 minutes, calculate the absolute error with respect to the theoretical probability, and compare the two probabilities.

### Define the variable and frequency distribution

$$X = \text{delivery time, in minutes}$$

The total sample size is

$$n = 6 + 14 + 18 + 9 + 3 = 50.$$

### C6.1 - Find and compare probabilities

The interval from 20 to 40 minutes contains the classes

$$20\text{--}30 \quad \text{and} \quad 30\text{--}40.$$

Their combined frequency is

$$k = 14 + 18 = 32.$$

Therefore, the estimated empirical probability is

$$\hat{P}(20 \leq X \leq 40) = \frac{32}{50} = 0.64.$$

The theoretical probability is

$$P(20 \leq X \leq 40) = 0.4866.$$

The absolute error is

$$E = |0.64 - 0.4866| = 0.1534.$$

Thus,

$$\hat{P} = 0.6400, \quad P = 0.4866, \quad E = 0.1534$$

The empirical estimate differs from the theoretical probability. Because the data are grouped, the empirical probability is based on class frequencies rather than individual observations.

## 2.4 Investment returns

A financial analyst records the percentage return of an investment over 10 randomly selected days:

$$2.1 \quad 3.4 \quad 1.8 \quad 4.2 \quad 2.7 \quad 3.1 \quad 1.5 \quad 2.9 \quad 4.0 \quad 2.4$$

The theoretical model for the daily return has mean

$$\mu = 2.8$$

and variance

$$\sigma^2 = 0.72.$$

**Question.**

**C4** Calculate the sample mean, sample variance, and sample standard deviation, and compare them with the corresponding theoretical values.

Define the variable and sample

$X$  = daily percentage return of the investment

The sample size is

$$n = 10 \text{ .}$$

**C6.2 - Calculate sample statistics**

The sample mean is

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i.$$

The sample variance is

$$s^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2.$$

The sample standard deviation is

$$s = \sqrt{s^2}.$$

Substituting the observations,

$$\bar{x} = \frac{2.1 + 3.4 + 1.8 + 4.2 + 2.7 + 3.1 + 1.5 + 2.9 + 4.0 + 2.4}{10}$$

$$\bar{x} = 2.81$$

The sample variance is

$$s^2 = \frac{1}{9} \sum_{i=1}^{10} (x_i - 2.81)^2$$

$$s^2 \approx 0.78$$

and therefore

$$s \approx 0.88 \text{ .}$$

The theoretical and sample statistics can be compared as

$$\mu = 2.80 \approx \bar{x} = 2.81$$

and

$$\sigma^2 = 0.72 \approx s^2 = 0.78, \quad \sigma \approx 0.85 \approx s = 0.88$$

The sample statistics are close to the theoretical parameters, but they are not identical because the sample is finite and random.

## 2.5 Daily energy consumption

A production facility records its daily electricity consumption, measured in MWh. The observations are summarized by the following frequency distribution:

|       |   |   |    |    |    |    |
|-------|---|---|----|----|----|----|
| $x_i$ | 8 | 9 | 10 | 11 | 12 | 13 |
| $f_i$ | 2 | 4 | 7  | 8  | 6  | 3  |

**Question.**

**C4** Calculate the sample mean, sample variance, and sample standard deviation of the daily electricity consumption.

**Define the variable and frequency distribution**

$X$  = daily electricity consumption, in MWh

The total sample size is

$$n = \sum_i f_i.$$

### C6.2 - Calculate sample statistics

For frequency data, the sample mean is

$$\bar{x} = \frac{\sum_i f_i x_i}{n}.$$

The sample variance is

$$s^2 = \frac{\sum_i f_i (x_i - \bar{x})^2}{n - 1}.$$

The sample standard deviation is

$$s = \sqrt{s^2}.$$

Using the frequency distribution,

$$n = 2 + 4 + 7 + 8 + 6 + 3 = 30.$$

Therefore,

$$\bar{x} = \frac{2(8) + 4(9) + 7(10) + 8(11) + 6(12) + 3(13)}{30}.$$

$$\bar{x} = 10.77$$

Then,

$$s^2 = \frac{\sum_i f_i (x_i - \bar{x})^2}{29}$$

giving

$$s^2 \approx 1.84 \ .$$

Therefore,

$$s \approx 1.36 \ .$$

## 2.6 Delivery times

A delivery company records the time required to complete deliveries, measured in minutes. The observations are grouped into intervals:

| Delivery time (min) | Frequency |
|---------------------|-----------|
| 10–20               | 6         |
| 20–30               | 14        |
| 30–40               | 18        |
| 40–50               | 9         |
| 50–60               | 3         |

**Question.**

**C4** Estimate the sample mean, sample variance, and sample standard deviation using the grouped data.

**Define the variable and grouped distribution**

$$X = \text{delivery time, in minutes}$$

For each interval, the midpoint is used to represent the observations in that class:

$$m_i = \frac{a_i + b_i}{2}.$$

Thus,

| Interval | Midpoint $m_i$ | Frequency $f_i$ |
|----------|----------------|-----------------|
| 10–20    | 15             | 6               |
| 20–30    | 25             | 14              |
| 30–40    | 35             | 18              |
| 40–50    | 45             | 9               |
| 50–60    | 55             | 3               |

The total sample size is

$$n = \sum_i f_i = 50.$$

## C6.2 - Calculate sample statistics

The grouped-data mean is

$$\bar{x} = \frac{\sum_i f_i m_i}{n}.$$

Therefore,

$$\bar{x} = \frac{6(15) + 14(25) + 18(35) + 9(45) + 3(55)}{50}.$$

$$\bar{x} = 32.8 \text{ min}$$

The grouped-data variance is

$$s^2 = \frac{\sum_i f_i (m_i - \bar{x})^2}{n - 1}.$$

Thus,

$$s^2 \approx 105.47 \text{ min}^2$$

and

$$s = \sqrt{s^2}.$$

$$s \approx 10.27 \text{ min}$$

These values are estimates because each observation within a class is represented by the class midpoint.